



A2 Nucleic acids

A · Unity and diversity · IB Diploma · Concept guide and original exam-style practice

Understand the concept

DNA and RNA are polymers of nucleotides, each built from a pentose sugar, a phosphate group, and a nitrogenous base (purines adenine and guanine; pyrimidines cytosine, thymine and uracil). DNA is double-stranded and antiparallel, held together by hydrogen bonds between complementary base pairs (A-T, C-G) and by a sugar-phosphate backbone, giving a stable structure suited to long-term storage and accurate replication of genetic information. RNA is usually single-stranded, uses ribose instead of deoxyribose, and uracil instead of thymine; its different forms (mRNA, tRNA, rRNA) carry out gene expression.

Worked example: follow the reasoning

Given the DNA template strand 5'-ATG CGT TAC-3', find the mRNA transcript. Step 1: RNA polymerase reads the template 3'→5' and builds mRNA 5'→3', pairing complementary bases. Step 2: pair each template base (A-T, T-A, G-C, C-G, G-C, T-A, T-A, A-T, C-G) to get, in the same left-to-right order as the template but representing the new strand read 3'→5': T A C G C A A T G. Step 3: reverse this to write the new strand 5'→3', and replace T with U (RNA): the mRNA is 5'-GUA ACG CAU-3'. Conclusion: the mRNA is complementary and antiparallel to the DNA template, with uracil replacing thymine.

Quick check · multiple choice

Which pair of bases forms complementary base pairs held by hydrogen bonds in a DNA double helix?

- A Adenine-Uracil and Guanine-Cytosine
- B Adenine-Thymine and Guanine-Cytosine
- C Adenine-Guanine and Thymine-Cytosine
- D Adenine-Cytosine and Thymine-Guanine

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A double-stranded DNA molecule contains 8000 base pairs. Analysis shows that 35% of all the bases in the molecule are adenine. [7]
 - (a) State the percentage of the bases that are thymine, and explain your reasoning. [2]
 - (b) Calculate the number of guanine bases in the molecule. [3]
 - (c) Explain one structural reason why DNA, rather than RNA, is the molecule used for long-term storage of genetic information in most organisms. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) 35% are thymine, because in double-stranded DNA adenine always pairs with thymine (complementary base pairing), so $\%A = \%T$.
- (b) $\%A + \%T = 70\%$, so $\%G + \%C = 30\%$, and since $\%G = \%C$, $\%G = 15\%$. Total bases = 8000 bp $\times 2 = 16\,000$ bases. Number of guanine bases = $15\% \times 16\,000 = 2400$.
- (c) DNA is double-stranded, so damage to one strand can be repaired using the other as a template, and the double helix with hydrogen-bonded base pairs and sugar-phosphate backbone is more chemically stable than the single-stranded, more reactive RNA, making DNA better suited to long-term, faithful storage of genetic information.

Quick check answer

B. In DNA, adenine always pairs with thymine (2 hydrogen bonds) and guanine always pairs with cytosine (3 hydrogen bonds).

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.